

STEPCONE-2025

Standard Operating Procedure (SOP)

Name of the Event: JAL-VEGA

Conducted by: Department of MECHANICAL

Duration of Event: 3 hours

Expected No of Registrations: 30 teams

Registration Fee: 1000/-

Date of the Event: 24-01-2025(Tentative)

About the Event:

Jal-Vega involves creating rockets using water as a propellant. These rockets typically consist of a plastic bottle partially filled with water, pressurized air, and a launch system. When the air pressure builds up inside the bottle, it forces the water out through a nozzle, generating thrust and launching the rocket into the air.

It consists of three rounds.

Details of the event:

Round 1: Range test at certain pressure

- The rocket has to travel a minimum cutoff distance of 50 m according to the pressure given. The projection angle is up to the participants.
- The points will be considered according to the following relation:
- Total Points = Range (in metres)/5.

Round2: Reaching the target

- Tests the stability and accuracy of the rocket towards the target.
- Two chances will be provided for launching the rocket.
- Scores will be based on the Target zone on which the rocket lands.
- Total Points = p(round 1) + (time Taken).

Round 3: Duration Challenge

- Rockets designed to achieve the longest possible flight duration.
- This often involves balancing the rocket's design to reduce air resistance and extend flight time.
- Scores will be allocated according to the judges.

Venue	Duration	Faculty Coordinator	Student Coordinator
Nearby Sports Arena (Helipad)	3 Hours	Dr. U Sudhakar Dr. S Ravi Babu	G. Divya N. Krishna veni

Guidelines of the Event:

- **Maximum of 3 participants in each team.** Individual participation is also allowed.
- All energy given to the rocket must come only from water pressure combination. No other source of energy is allowed.
- The teams will be disqualified/given another chance to remodel if their model is found to be dangerous in any way. No glass or sharp objects should be used in making the rockets.

- Jal-Vega have multiple stages. Parachute, gliding and other mechanical launch mechanisms are allowed. Booster mechanisms are allowed as well.
- You are only allowed to use plastic bottles specifically designed for holding pressure, or that have been pressure tested (for example carbonated-drink bottles).
- More than one bottle can be used but the maximum capacity of the bottle/bottles should be 1L – 1.5 L. It is not necessary for both the bottles to be identical.
- Only water should be used as the working fluid. No other fluid is allowed. Participants should only use the water provided by organizers.
- Foot pumps will be provided by the organizers, this pump is compatible with standard bike valves. Participants must only use the pump provided by the organizers.
- Participants can make different models for different rounds but the volume and the bottles provided should be the same.
- Use of electronic components and chemical explosives is banned.

Evaluation and result:

The judges will evaluate the participants performance in Each round and the final result will be determined based on the marks obtained by the participants in each Round.